

Pathology of Bone and Joints – Part 1

Principles and Practices of Osteopathic Medicine I

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Do.
Make.
Heal.
Innovate.

Reinvent the Future.

OUTLINE

Osteoarthritis
Rheumatoid Arthritis
Achondroplasia
Osteogenesis Imperfecta
Osteoporosis
Osteopetrosis
Osteomalacia / Rickets
Paget Disease of Bone (osteitis deformans)
Osteonecrosis (avascular necrosis)



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OSTEOARTHRITIS – DEGENERATIVE JOINT DISEASE

Cartilage Degeneration



Synovial Joints Failure
(structural and functional)

- Most common type of joint disease

PRIMARY-IDIOPATHIC: without apparent initiating cause - oligoarticular but may be generalized

SECONDARY: About 5% of cases, in younger people with predisposing conditions

OSTEOARTHRITIS – DEGENERATIVE JOINT DISEASE

ARTICULAR CARTILAGE: → DEGENERATION



DISORDERED REPAIR

Biomechanical stress: principal pathogenic mechanism

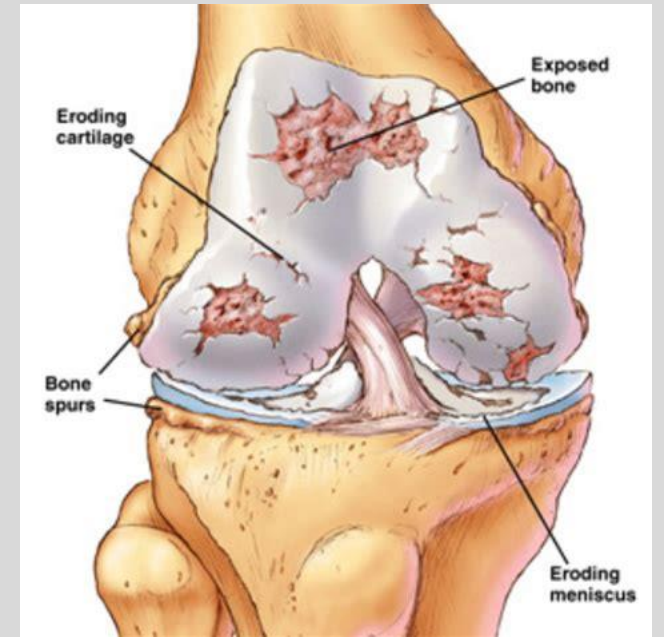
Genetic factors:

- polymorphisms in genes encoding components of matrix and signaling molecules



CHONDROCYTE INJURY: MATRIX ALTERATION

- **Chondrocytes** proliferate and continuously synthesize proteoglycans (PG)
- **OA:** degradation exceeds synthesis of PG

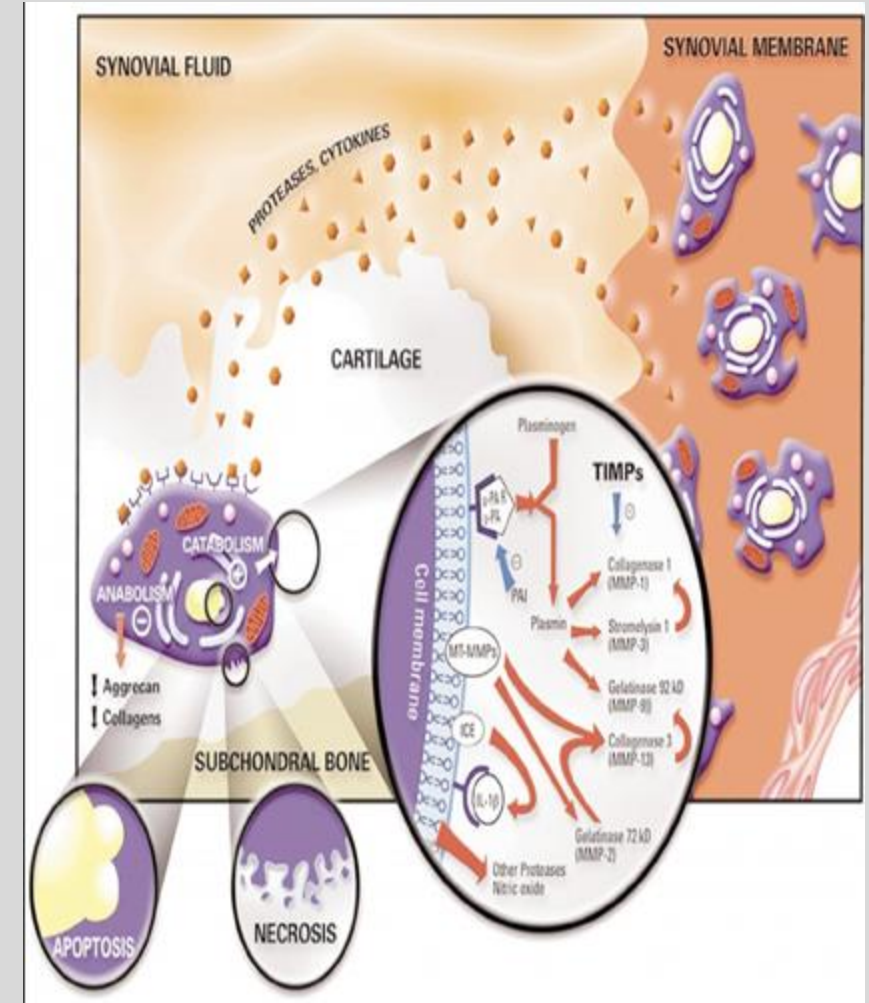


OSTEOARTHRITIS

- Cartilage: Chondrocytes also secrete matrix metalloproteases (**MMPs**): degrade type II collagen network
- Tissue inhibitor of metalloproteinase (**TIMPs**) counteract proteolytic enzymes produced by chondrocytes (Imbalance of MMPs and TIMPs demonstrated in OA)
- Cytokines and diffusible factors from chondrocytes and synovial cells (especially TGF- β which induces production of MMPs), TNF, prostaglandins, and nitric oxide, also implicated in OA
- **Chronic, low-level inflammation** contributes to disease progression
- Advanced disease: **chondrocyte loss and matrix degradation**

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OSTEOARTHRITIS EVOLUTION

Early Stages

- Chondrocytes proliferate: clusters
- Matrix water content increases,
- Proteoglycan concentration decreases,
- Collagen type II fibers are cleaved



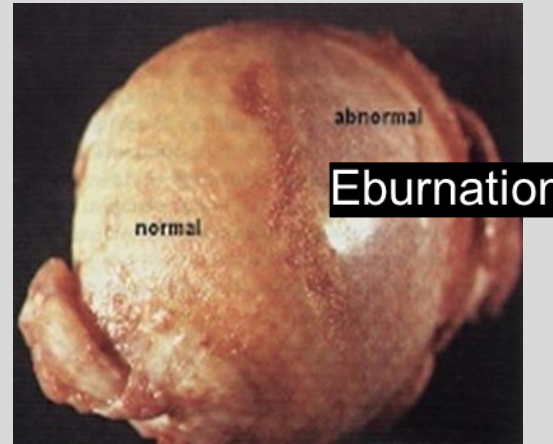
FISSURES AND CLEFTS

Articular surface: granular and soft

Chondrocytes die: cartilage is sloughed into joint



LOOSE BODIES



Exposed subchondral bone : **NEW ARTICULAR SURFACE**, polished by friction with opposing surface → polished ivory (bone eburnation) aspect

Articular bone : sclerosis, develops small fractures, creating gaps → **CYSTS** : synovial fluid

OSTEOPHYTES: Mushroom-shaped bony outgrowths develop at the edge (covered by fibrocartilage and hyaline cartilage which will gradually ossify)

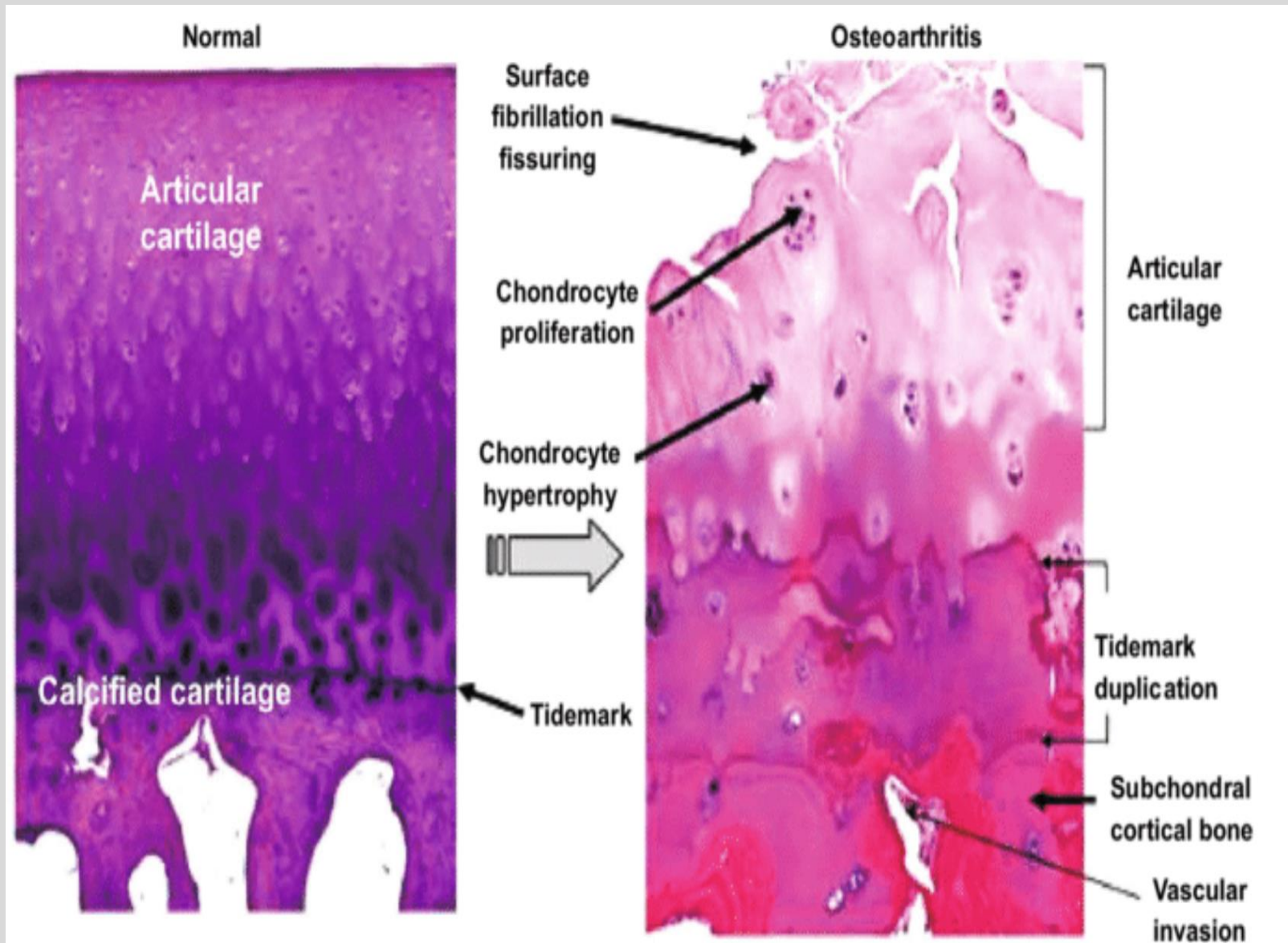
Synovium usually only mildly congested and fibrotic, contains few inflammatory cells

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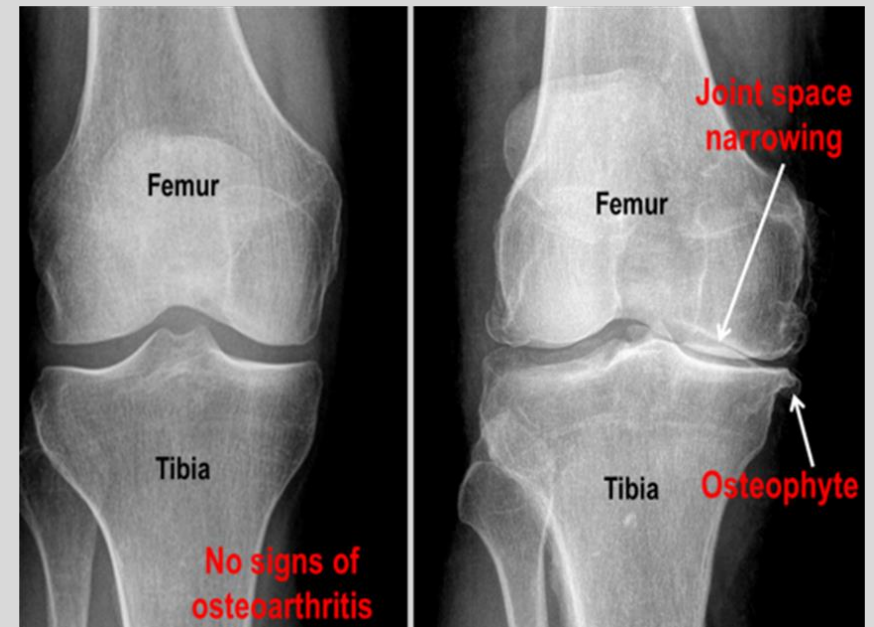
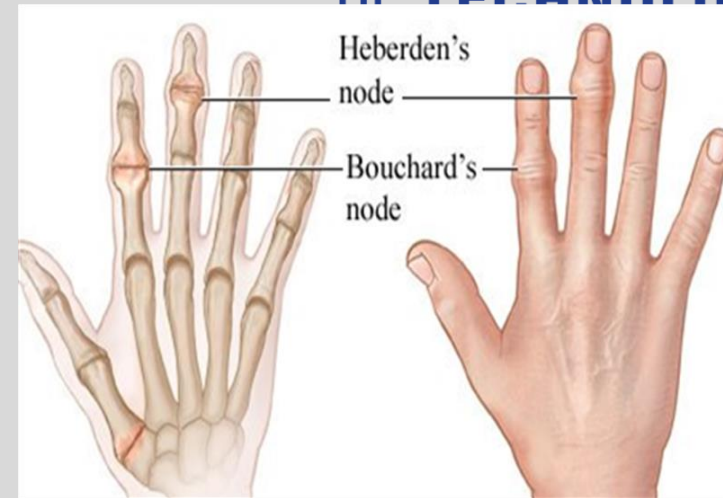
OSTEOARTHRITIS

- **Location: *Asymmetrically*** involves weight bearing joints
 - Usually, only one or few joints involved, especially hips, knees, lower lumbar and cervical vertebrae, proximal and distal interphalangeal joints of fingers, first carpometacarpal joints, and first tarsometatarsal joints
- **Joint findings:**
 - **Osteophytes (bone spurs)** – Response to stress
 - **“Joint Mice”** dislodged cartilage and subchondral bone in joint
 - Joint space narrowing
 - Subchondral sclerosis (**eburnation** – ivory-like polished surface)
 - **Non-systemic process** (WBC <2000) on aspiration
 - Osteophytes=Nodes

Distal Interphalangeal (DIP) – HeberDen nodes
Proximal Interphalangeal (PIP) – Bouchard nodes

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athic



RHEUMATOID ARTHRITIS

- Chronic autoimmune inflammatory **systemic** disorder
- The **JOINTS** show a nonsuppurative proliferative and inflammatory **SYNOVITIS**



PANNUS: GRANULATION TISSUE



ARTICULAR CARTILAGE AND BONE EROSION



**ARTICULAR CARTILAGE DESTRUCTION
JOINT FUSION(ANKYLOSIS)**

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Extra-articular lesions: skin, heart,
blood vessels, and lungs

Prevalence in US: 1%;
Incidence in the US: peaks second to
fourth decades; 3X more common in
women

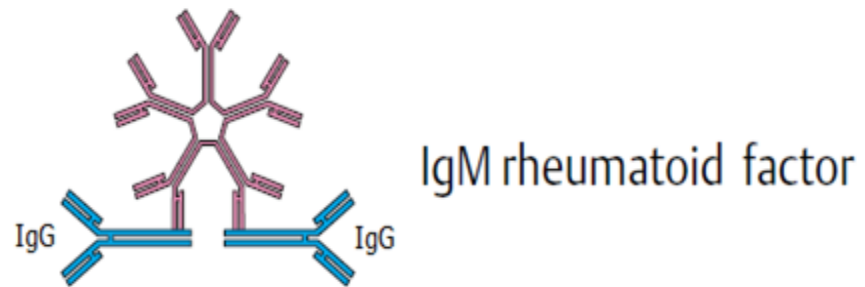
- Autoimmune disease **induced by CD4+ helper T cells** (potentially after contact with arthritogenic agent, microbial or self-antigen)
- **Genetic predisposition and environmental factors**
- Pathologic changes in RA mediated by antibodies against self antigens and inflammation induced by cytokines, mostly secreted by **CD4+ T cells**
- Activated CD4+ T cells release cytokines: stimulate other inflammatory cells, leading to tissue injury:
 - • IFN- γ from Th1 cells: activates macrophages and synovial cells
 - • IL-17 from Th17 cells: recruits neutrophils and monocytes
 - • RANKL, expressed on activated T cells: stimulates bone resorption
 - • TNF and IL-1 from macrophages: stimulate synovial cells to secrete proteases that destroy hyaline cartilage.
- **TNF most firmly implicated in RA pathogenesis (anti-TNF biologics in treatment, see later)**

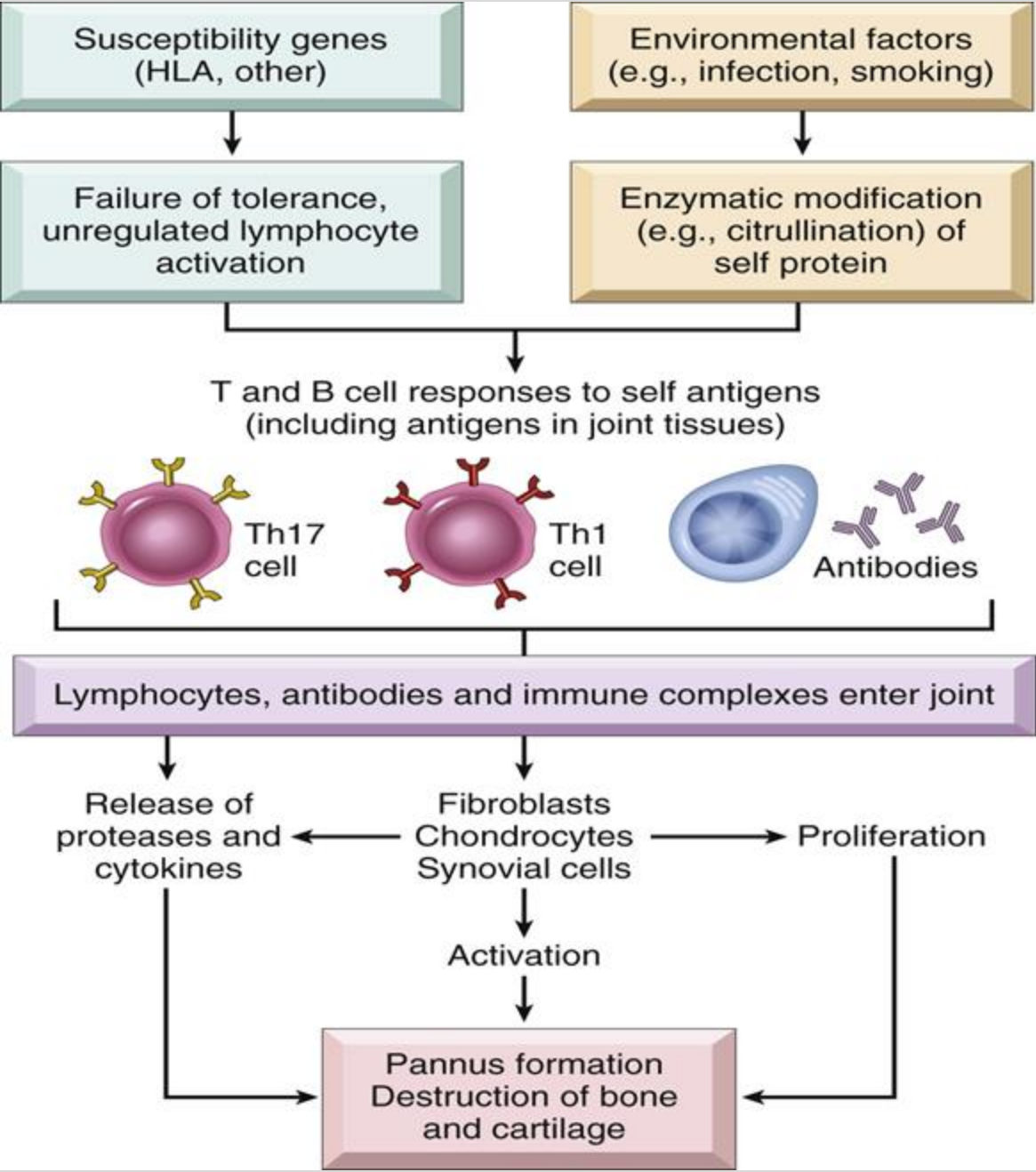
RHEUMATOID ARTHRITIS

- Synovium often contains **germinal centers with secondary follicles and abundant plasma cells**, some secrete antibodies that react with self antigens
 - These autoantibodies (may be synthesized in lymphoid organs), are specific for citrullinated peptides (CCPs)
- A. Anti-citrullinated peptide antibodies (ACPAs):** diagnostic markers, detected in serum of up to 70% of RA patients

B. IgM and IgA autoantibodies that bind to IgG Fc regions present in 80% of cases - known as **rheumatoid factor:** may be deposited in joints as immune complexes; **not present in all patients, can also be found in patients without RA**

- **Anti-Cyclic Citrullinated Peptide Antibody (Anti-CCP): More specific than rheumatoid factor for RA**
- About 50% of risk of developing RA related to inherited genetic susceptibility: HLA-DR4 allele associated with ACPA-positive RA

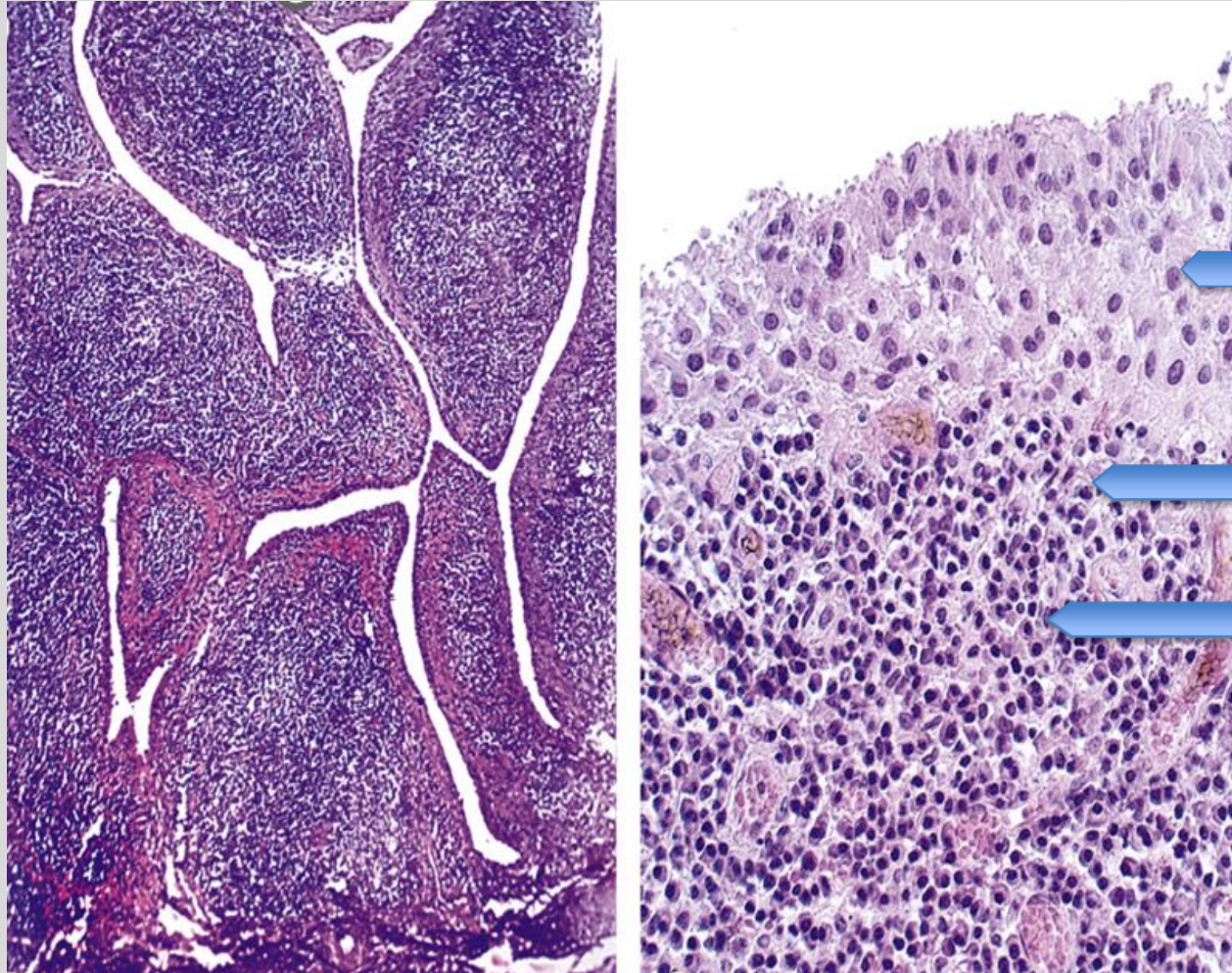




RHEUMATOID ARTHRITIS

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Histopathology: PANNUS

mass of edematous synovium

inflammatory cells

granulation tissue and fibroblasts

It grows over and causes **erosion** of articular cartilage

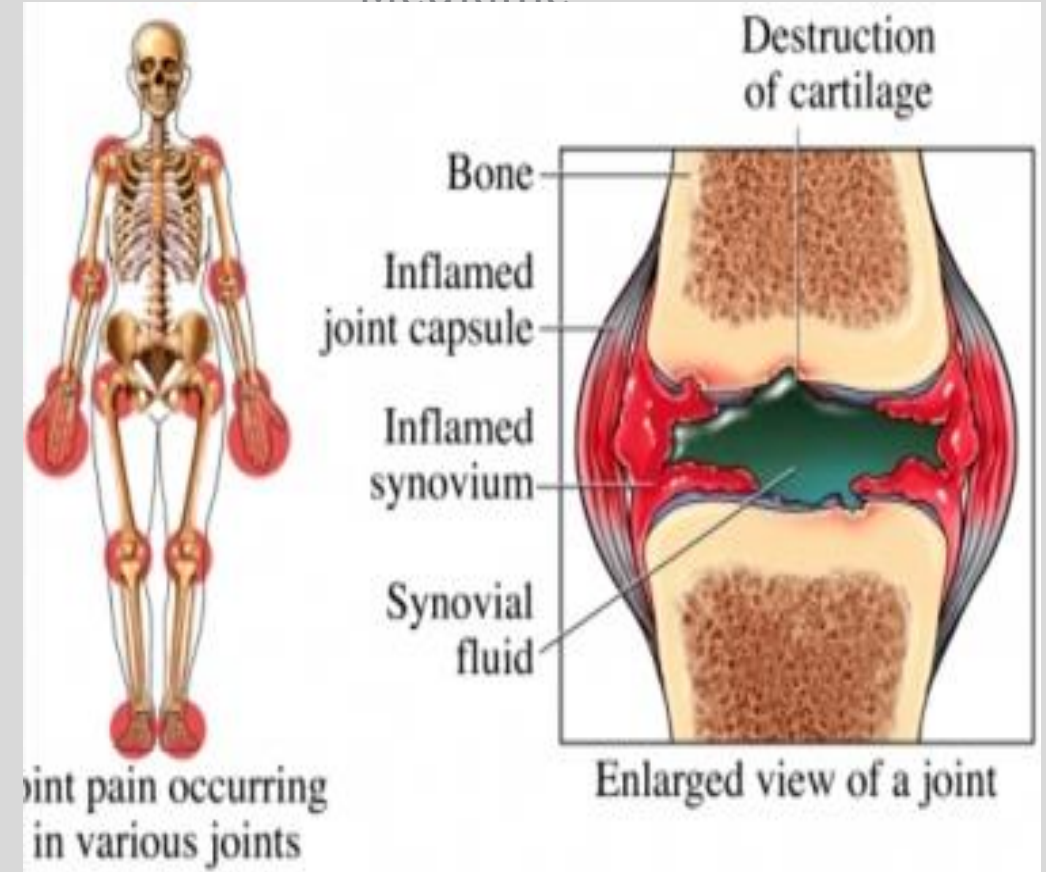
Pannus bridges determines a **fibrous ankylosis** which could **ossify** resulting in **bone fusion and bony ankylosis**

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- **Location:** *Symmetric* joint involvement
 - **Hands** (*DIP spared*), wrists, hips, knees, cervical spine, feet, ankles, elbows, shoulders
- **Pain:** usually worse in **morning**
- **Systemic symptoms;** fever, fatigue, weight loss
- **Labs**
 - Anti-CCP (most sensitive and specific test)
 - Elevated ESR
 - Elevated CRP
 - Positive RF titer
 - Joint fluid testing (decreased complement)



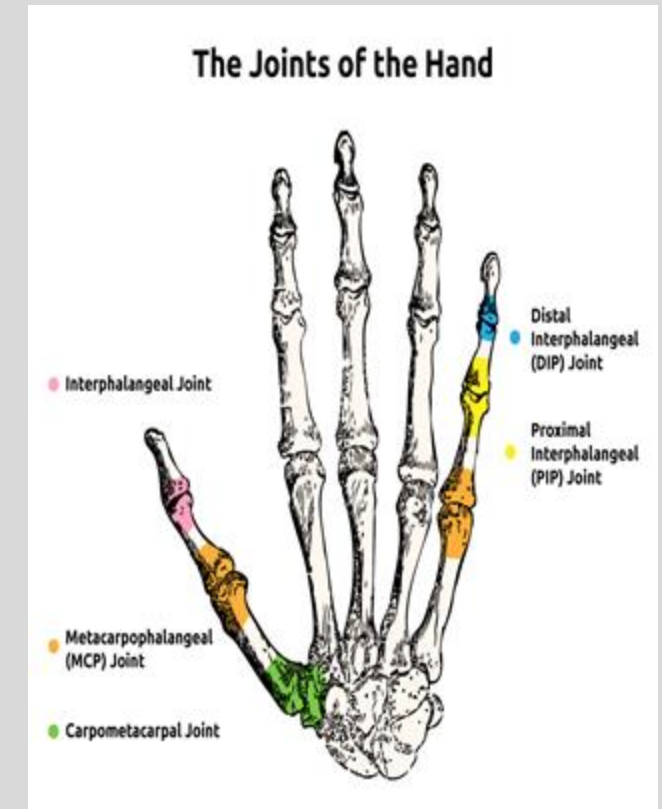
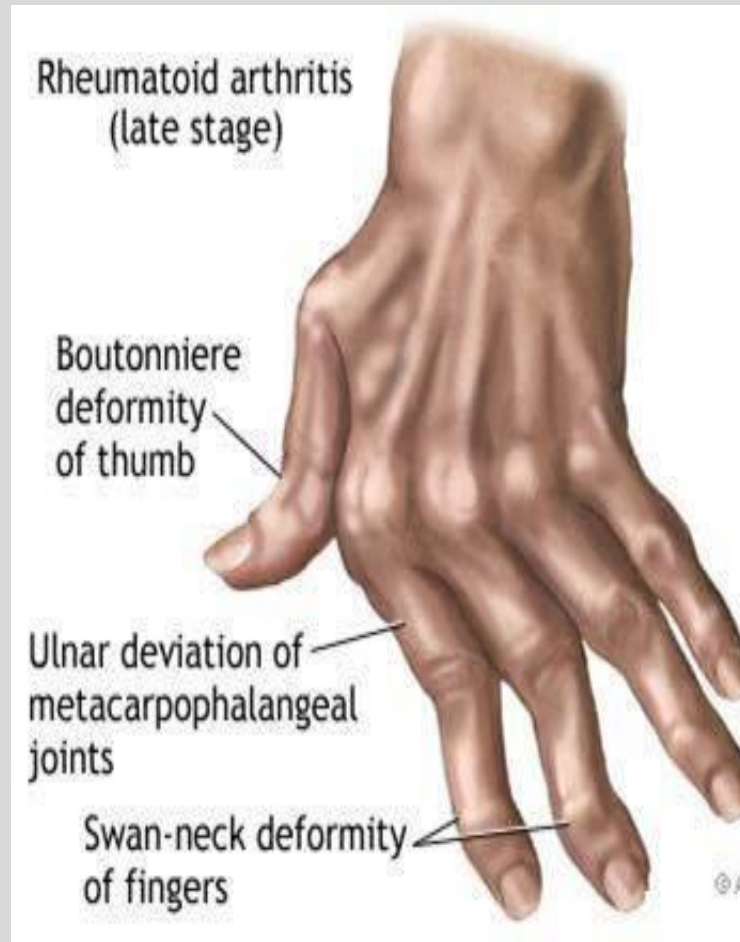
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- **Hand findings:**

- Proximal Interphalangeal (PIP) – **Bouchard nodes**
- **RA DOES NOT include Heberden Node (DIP)**
- Inflammation in tendons, ligaments, sometimes adjacent skeletal muscle produces:
 - **Radial deviation of wrist**
 - **Ulnar deviation of fingers**
 - **Flexion-hyperextension fingers (swan-neck and boutonnière deformities)**



<https://www.wristsupports.co.uk/blog/what-is-cmc-instability.ht>

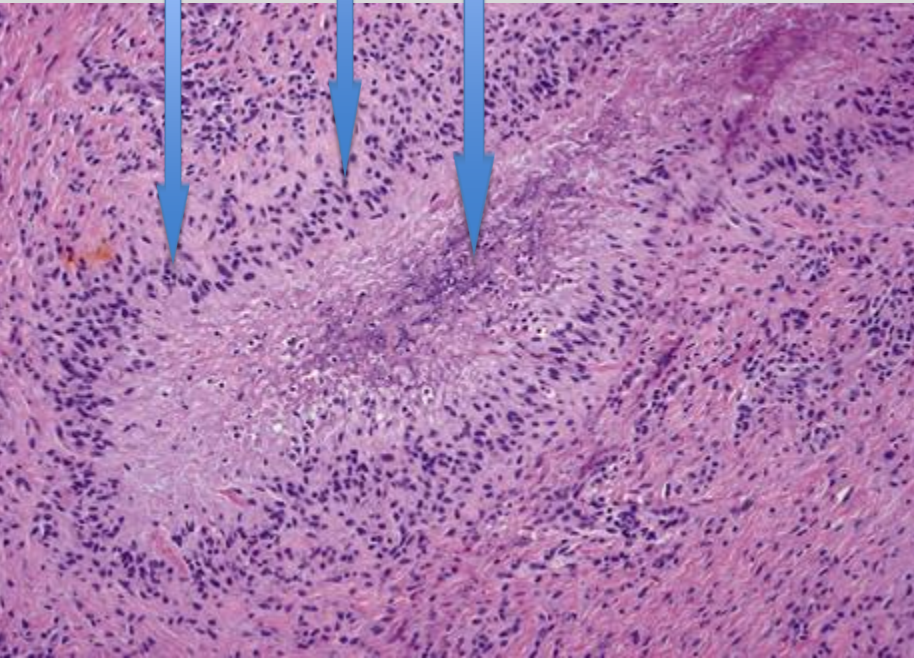
RA: EXTRA-ARTICULAR FEATURES

- **Rheumatoid nodules** - in subcutaneous tissue of forearm, elbows, occiput, and lumbosacral area (pressure areas): small firm, nontender, round to oval masses;
 - in lungs (Caplan syndrome)
 - Micro: necrotizing granulomas with central zone of fibrinoid necrosis surrounded by prominent rim of palisading, activated macrophages, many lymphocytes and plasma cells



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- **Acute necrotizing vasculitis of small and large arteries:** pleura, pericardium, or lung; produces **purpura, cutaneous ulcers, and nail bed infarction**
- **Uveitis, scleritis and keratoconjunctivitis**
- **Interstitial lung disease, pleuritic**
- **Pericarditis**
- **Anemia of chronic disease**
- **Felty syndrome: neutropenia + splenomegaly**

	Osteoarthritis	Rheumatoid Arthritis
Path	Mechanical / Degenerative Cartilage degeneration and fragmentation, bone spurs, subchondral cysts; minimal inflammation	Autoimmune Inflammatory pannus invading and destroying cartilage; severe chronic inflammation; joint fusion (ankylosis)
Role of Inflammation	May be secondary; inflammatory mediators exacerbate cartilage damage	Primary: cartilage destruction is caused by T cells and antibodies reactive with joint antigens
Timing	Worse with activity (mechanical)	Worse upon waking (build up of inflammatory factors) – pain <i>improves</i> with movement
Location	Asymmetric, weight bearing joints (Hips and Knees) Hands: DIP, PIP, CMC	Symmetric, Hands and Feet , knees, cervical spine Hands: PIP, MCP (ulnar deviation), (<i>DIP spared</i>), swan neck deformity, boutonniere deformity
Labs	Non-inflammatory aspirate <2000WBC No serum antibodies	Increased; Anti-CCP, RF, ESR, CRP. Decreased complement Various antibodies, including ACPA, RF
Joint Findings	Joint space narrowing, Osteophytes, joint mice, subchondral sclerosis (eburnation)	Pannus formation , juxta-articular osteopenia, Soft tissue swelling, subchondral cysts, joint space narrowing
Extra-Articular	None	Rheumatoid Nodules (fibrinoid necrosis with palisading histiocytes), pleuritis, pericarditis, Felty syndrome

JUVENILE IDIOPATHIC ARTHRITIS

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- Heterogeneous group of disorders, unknown cause: present with **arthritis before 16 years old and last at least 6 weeks**
- Leukocytosis, Thrombocytosis, Anemia, Antinuclear antibody (ANA) seropositivity; Increased ESR and CRP
- **Pathogenesis** undefined; associations with certain HLA and PTPN22 gene variants
- Daily spiking fevers, salmon pink macular rash, arthritis (+2 joints)
- Anterior uveitis
- Rheumatoid nodules usually absent
- Long-term JIA prognosis variable; many have chronic disease; about 10% develop serious functional disability
- **Treatment:** NSAIDs, steroids, methotrexate, TNF inhibitors

DEFECTS IN HORMONES AND SIGNAL TRANSDUCTION PROTEINS

ACHONDROPLASIA

- **Most common skeletal dysplasia and major cause of dwarfism**

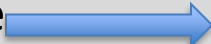
THANATOPHORIC DYSPLASIA

- **Most common lethal form of dwarfism**
- ~1 in every 20,000 live births –

Pathogenesis: FGFR3 gain-of-function mutations different than those that cause achondroplasia

- Disproportionately short (micromelic) limbs, frontal bossing, macrocephaly, small chest cavity, and bell-shaped abdomen
- Underdeveloped thoracic cavity determines respiratory insufficiency
- The patients die at birth

Histology:

Growth Plate  reduced chondrocyte proliferation and disorganization within zone of proliferation

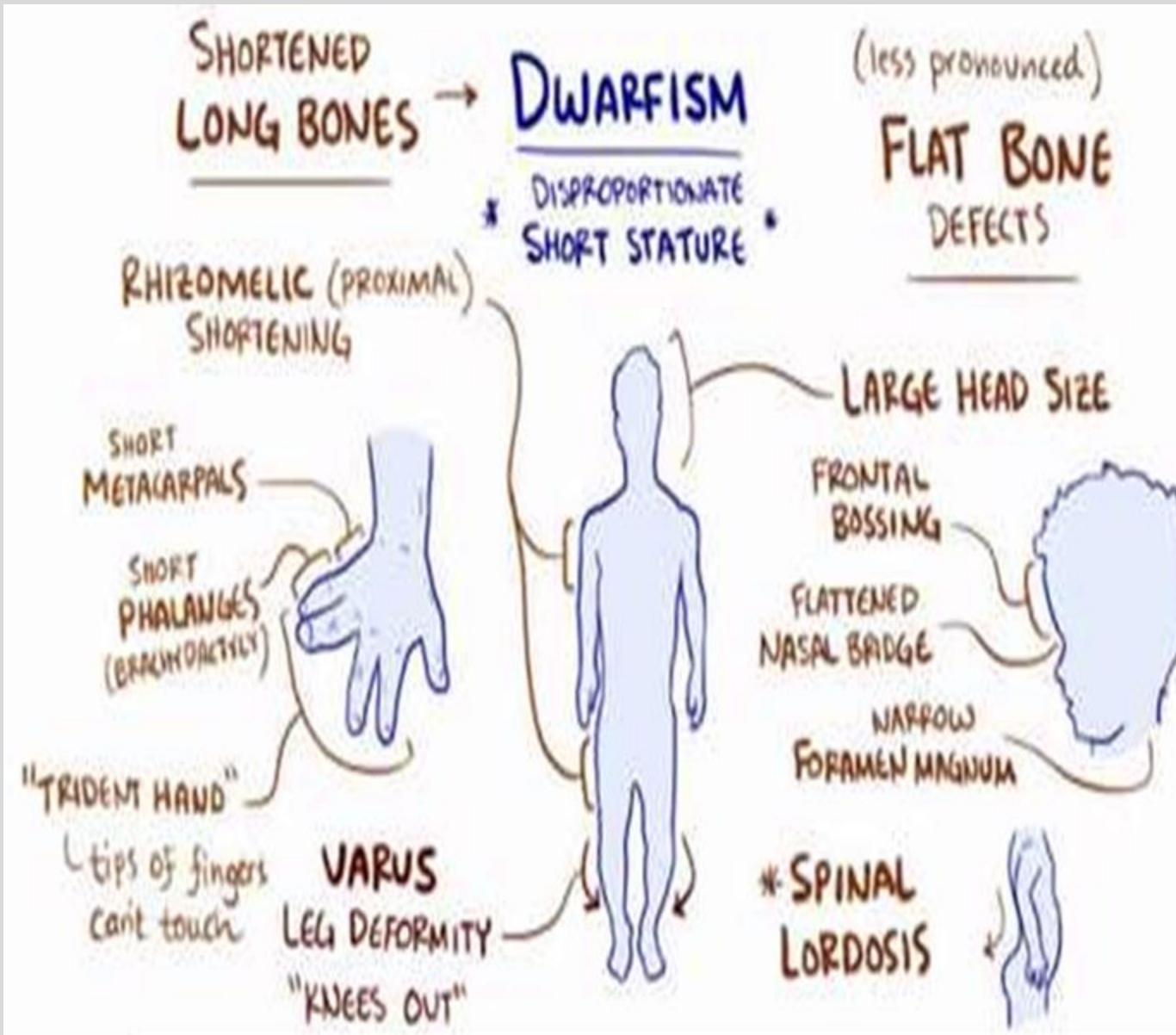
ACHONDROPLASIA – DWARFISM

- **Autosomal dominant** disorder: **gain-of-function mutations in the FGF receptor 3 (FGFR3) gene**
- FGF-mediated FGFR3 activation normally inhibits endochondral growth which is exaggerated by FGFR3 gain-of-function mutations
- **Retarded cartilage growth:** shortened proximal extremities, enlarged head with bulging forehead, depression of root of nose but trunk of relatively normal length
- Normal longevity, intelligence, and reproductive status
- 85% sporadic (increased risk with increased paternal age)

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- Constitutive activation of **FGFR3** (fibroblast growth factor receptor 3) leading to decreased chondrocyte activity
- Failure of endochondral ossification (longitudinal bone growth) and as a result the patients have short limbs
- Membranous ossification unaffected: Skull, facial bones, and clavicle **NORMAL** - large head and torso relative to limbs



OSTEOGENESIS IMPERFECTA— DEFECT IN EXTRACELLULAR PROTEINS

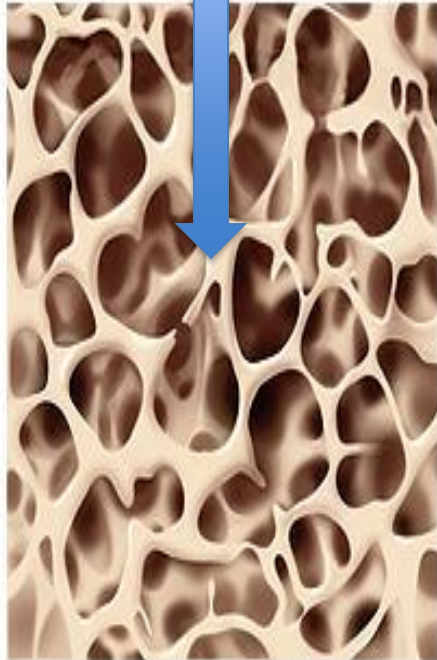
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Too little bone!



Magnified view of normal bone



Magnified view of bone with
osteogenesis imperfecta

Type 1 Collagen Diseases (Osteogenesis Imperfecta) aka brittle bone disease

Most common inherited disorder of connective tissue, phenotypically heterogenous; caused by deficiencies in **type I collagen synthesis** in

- Bone and other tissues with type I collagen (joints, eyes, ears, skin, and teeth)

Pathogenesis Caused by **mutations in genes encoding the $\alpha 1$ and $\alpha 2$ chains of type I collagen**; mutations result in misfolding of collagen polypeptides and defective assembly of higher-order collagen chains - **mostly autosomal dominant COL1A1 or COL1A2**

OSTEOGENESIS IMPERFECTA

- Present with fractures at birth
- In OI we may see mutations that result in decreased synthesis of qualitatively normal collagen associated with **mild skeletal abnormalities**
 - Or more **severe or lethal phenotypes** associated with mutant collagens that interfere with triple helix formation
- **Type 1 OI:** normal life span with childhood fractures that decrease in frequency following puberty
- **Type 2OI:** **uniformly fatal** in utero or during perinatal period; severe bone fragility and multiple intrauterine fractures

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Clinical Features

- Bones – **Multiple fractures** due to minimal trauma (confused with child abuse)
- Ears – **Hearing loss** due to both sensorineural deficit and impaired conduction because of abnormalities in bones of middle and inner ear



- Eyes – **Blue sclerae**: decreased collagen content, makes sclera translucent with partial visualization of underlying choroidal veins



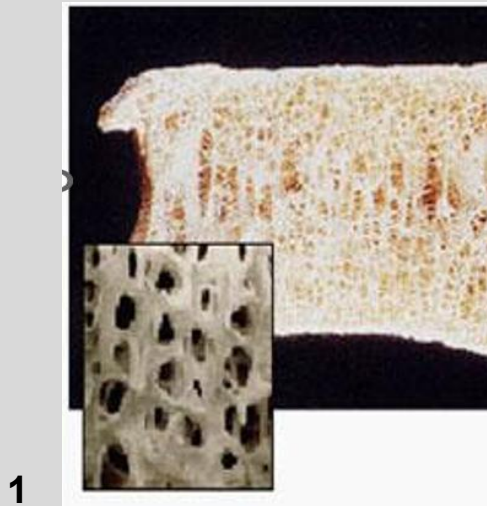
- Teeth - dental imperfections (**small, misshapen, and blue-yellow teeth**) secondary to dentin deficiency.

OSTEOPOROSIS

- Defined as osteopenia (decreased bone mass) severe enough to increase risk of fracture
- Xray: bone mass at least 2.5 standard deviations below mean peak bone mass in young adults
- **Most common forms of osteoporosis are senile and postmenopausal**

Senile osteoporosis:

- **A. Diminished ability to make bone** (reduced proliferative and biosynthetic capacity and response to growth factors of osteoblasts)
- **B. Decreased physical activity** with normal aging



Post menopausal osteoporosis

Estrogen Deficiency: after menopause, up to 2% of cortical bone and 9% of cancellous bone may be lost per year.

- There is both bone formation and resorption, but **resorption dominates**
- There is increased secretion of IL-6, TNF, and IL-1 by innate immune cells in blood and marrow (unknown mechanisms)



Increases in RANK-L
Decrease in OPG

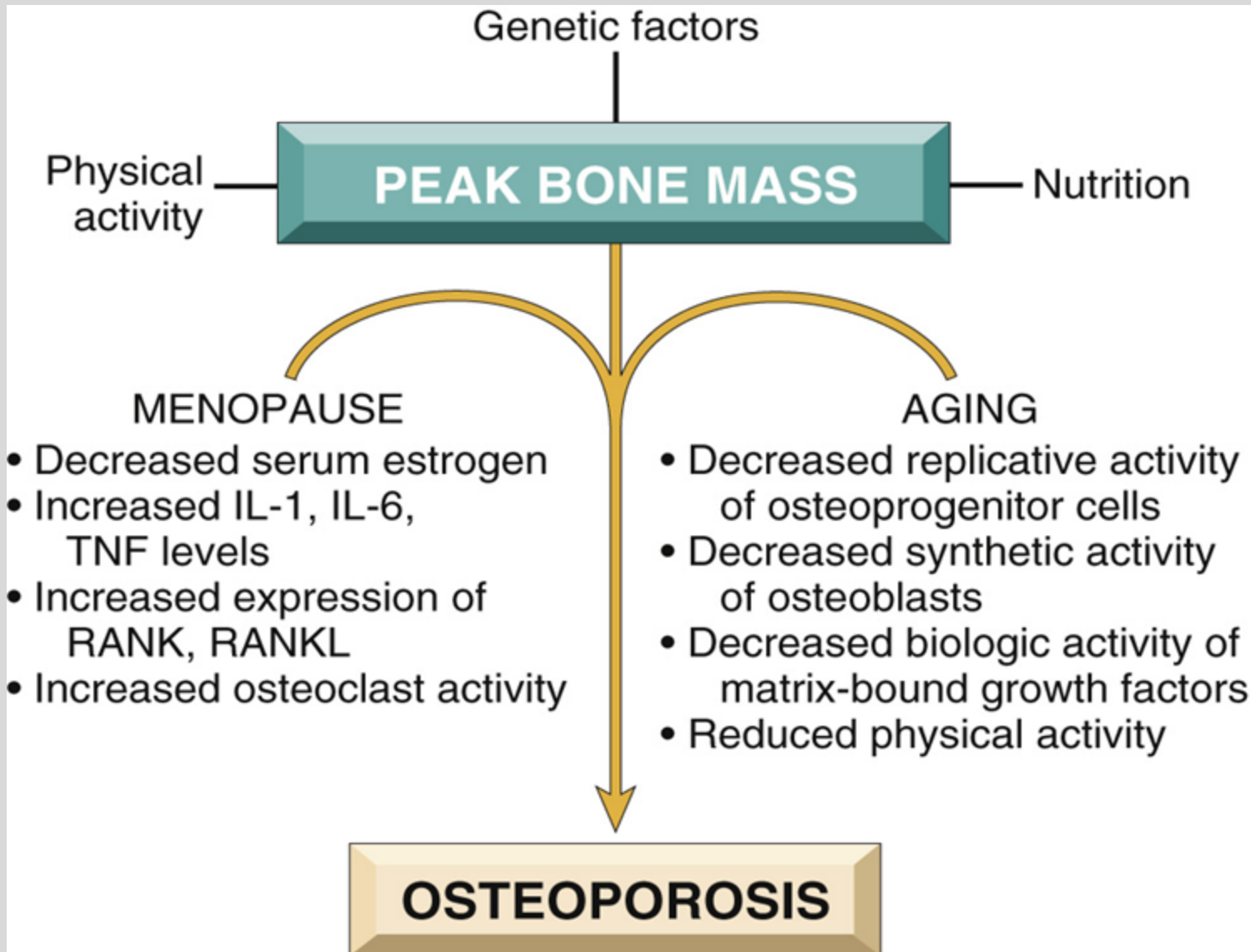


Stimulate osteoclast recruitment and activity

- **Histology: normal bone, decreased in quantity;** entire skeleton affected, certain bones more severely affected
- **Postmenopausal osteoporosis** affects mainly bones or portions of bones with increased surface area (e.g. vertebral bodies) - trabecular plates are perforated and thinned; they lose their interconnections and to microfractures eventually **vertebral collapse (vertebral compression)** would develop.

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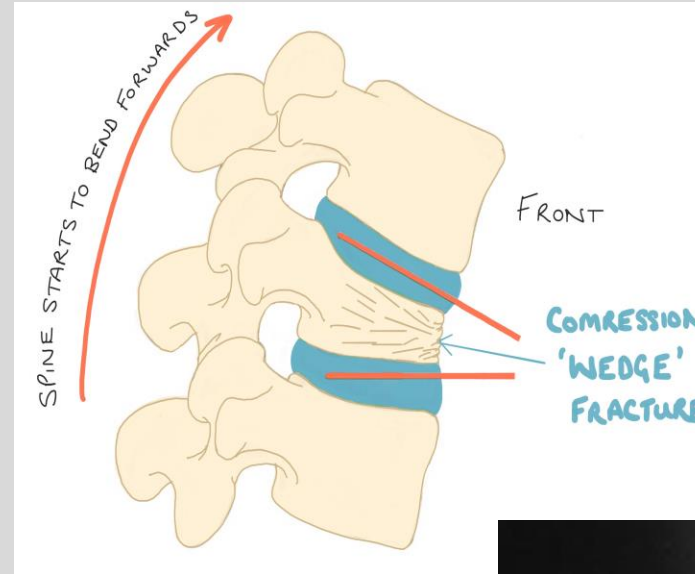
- Incidence: 10 million Americans and 200 million people worldwide have osteoporosis
- 1.5 million osteoporotic fractures occur each year
- **Male: female ratio is 1:4**
- Men have higher prevalence of secondary osteoporosis (60%)

- **Fractures:** Vertebral Body > Hip > Wrist

- Vertebral compression fracture: (esp. in thoracic and lumbar regions): painful, when multiple, can cause significant loss of height and deformities
 - lumbar lordosis and kyphoscoliosis

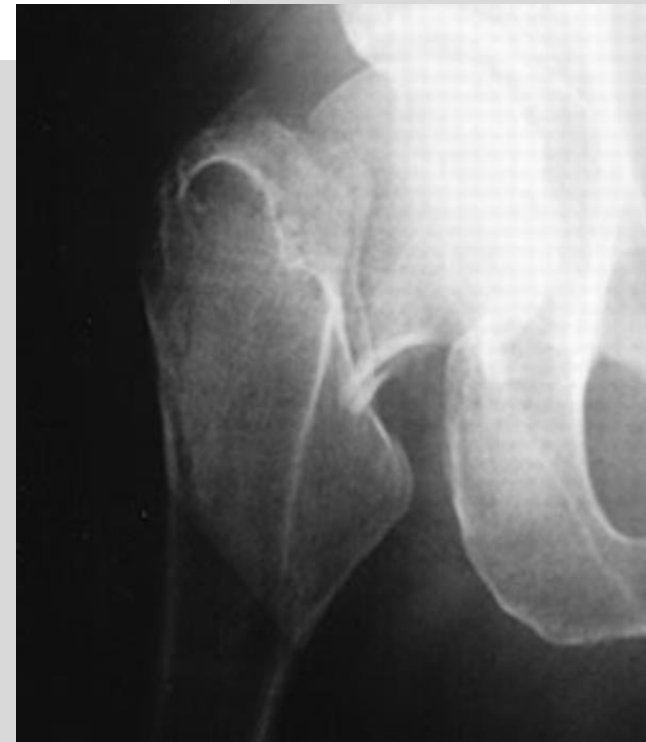
- Femoral neck fracture:
Leading to immobility

- Distal radius fracture



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- **Osteoporosis - detected in plain X-rays when 30% to 40% of bone mass loss**

- **Diagnosis:** Dual Energy X-ray Absorptiometry (DEXA scan) screening recommended for all women > 65 years
 - T-Score: Bone density compared to a healthy 30 year old
 - $\leq (-2.5)$: Osteoporosis
 - $(-2.5) - (-1.0)$: Osteopenia
 - $\geq (-1.0)$ Healthy bone
- **Prophylaxis/Treatment**
 - Diet (Ca, Vit D supplements)
 - Exercise – weight bearing
 - Bisphosphonates – reduce osteoclast activity

OSTEOPETROSIS

- Rare genetic diseases: **decreased bone resorption due to deficient osteoclast development or function**
- Diffuse, symmetric skeletal sclerosis
- Bones are brittle, fracture easily
- **Numerous variants:** variable inheritance and severity of clinical features
 - Most mutations in osteopetrosis **interfere with acidification of the osteoclast resorption pit** (needed for dissolution of calcium hydroxyapatite within the matrix)

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Mutations impair Carbonic anhydrase II: there is no H⁺ acidic environment necessary for osteoclast function

Defective osteoclasts: Can't resorb bone properly but the bone fractures easily.

Bone fills marrow space leading to **pancytopenia and extramedullary hematopoiesis**

X-ray: Symmetric sclerosis: the so-called bone in bone pattern

- Deposited bone not remodeled, tends to be woven rather than lamellar

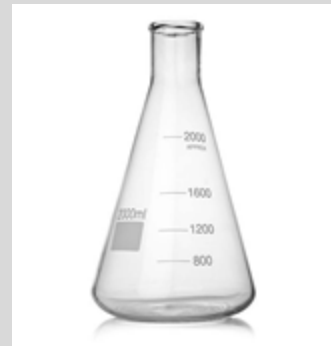
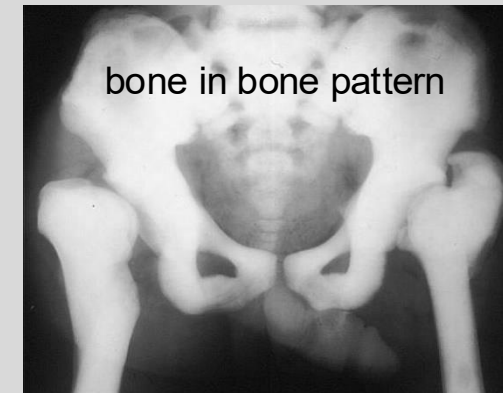
Erlenmeyer flask deformity – no medullary cavity (due to deficient osteoclast activity).

- Narrowed neural foramina and cranial nerve compression
- Palsies and deafness
- Increased risk of fracture: LE > UE > Axial Skeleton
- **AR:** Severe infantile osteopetrosis seen in utero or soon after birth
- **AD:** milder; may not be detected until adolescence

Treatment: Bone marrow transplant might be potentially curative

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OSTEOMALACIA & RICKETS

Pathogenesis:

- Metabolic – Defective mineralization of osteoid or cartilaginous growth plates
 - Low vitamin D, low sunlight, kidney disease

Osteomalacia - Adults: Defective mineralization of osteoid

Rickets - Kids: Defective mineralization of cartilaginous growth plates

Epidemiology

- 1/1000 in US –dairy product fortification with vitamin D and sunlight

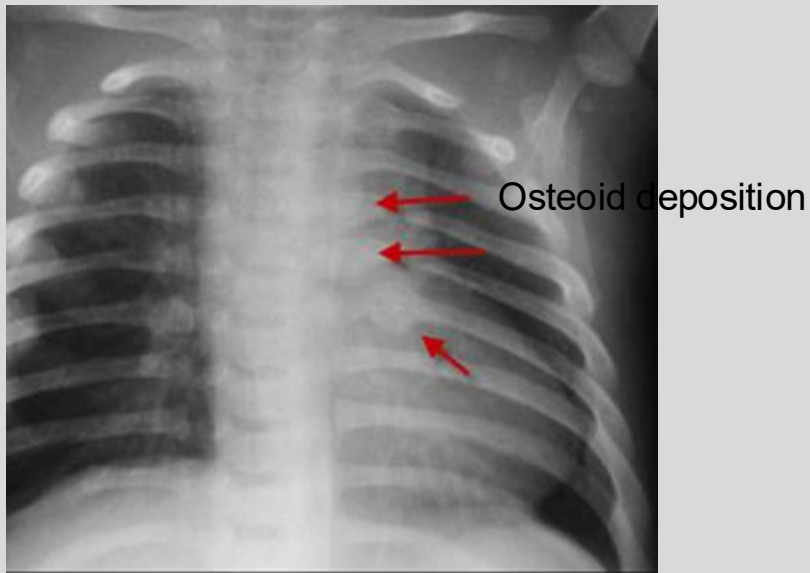
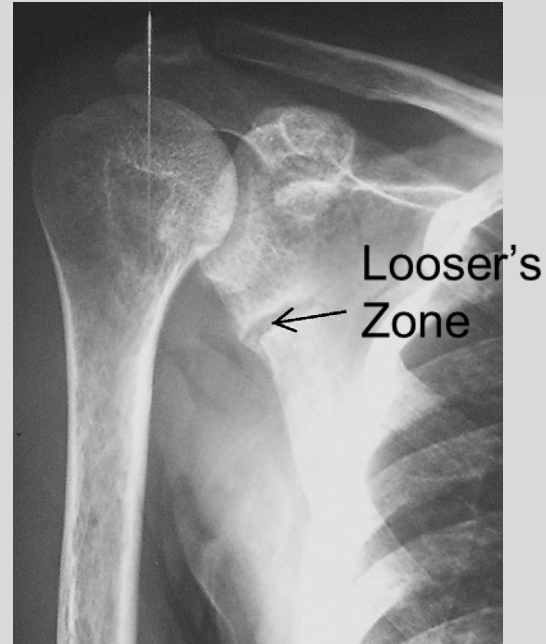
• Labs:

- ↓vitamin D: ↓Ca absorption: ↑PTH secretion: ↓ PO_4^{-3}
- Osteoblast hyperactivity ↑**ALK Phos**

- **Treatment:** Large doses of Vitamin D (1000 IU/day)

OSTEOMALACIA & RICKETS

- **Osteomalacia:** Looser's zone (insufficiency fracture) seen at the axillary border of the scapula, rib, or posterior ulna
- **Rickets:** epiphyseal widening and metaphyseal cupping/fraying
 - Genu varum – Bow legged
 - Rachitic rosary – Osteoid deposition at costochondral junction



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PAGET DISEASE

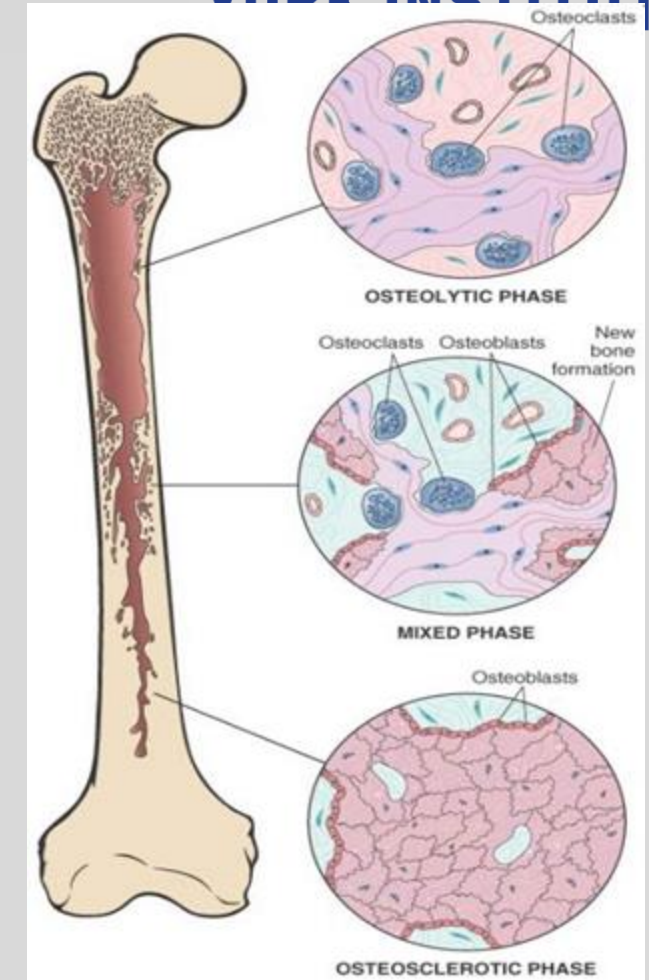
- **Increased**, but **disordered**, structurally unsound bone mass
- Three sequential phases:
 - (1) **osteolytic stage**
 - (2) **mixed osteoclastic-osteoblastic stage**
 - (3) **burned-out quiescent osteosclerotic stage**
- Presents in late adulthood and becomes progressively more common with increasing age
- Relatively common in **Caucasians** in England, France, Austria, regions of Germany, Australia, New Zealand, and US; rare in Scandinavia, China, Japan, and Africa
- Exact incidence not known because of **many asymptomatic patients**
- Affects about 1% of the US population > 40 years of age

PAGET DISEASE-OSTEITIS DEFORMANS

- Genetic and environmental contributions
- Approximately 50% of familial Paget disease and 10% of sporadic cases associated with mutations in the SQSTM1 gene which increases NF- κ B activity, enhancing osteoclast activity
- Chronic infection of osteoclast precursors by measles or other RNA viruses

Pathogenesis:

- **Osteolytic:** $\uparrow$ Osteoclastic activity – bone resorption
- **Mixed:** $\uparrow$ Osteoclast & Osteoblast activity – resorption and compensatory bone reformation
- **Osteosclerotic (“burnt out” phase):** $\uparrow$ Osteoblasts –bone formation
 - All 3 phases may co-exist in a given bone

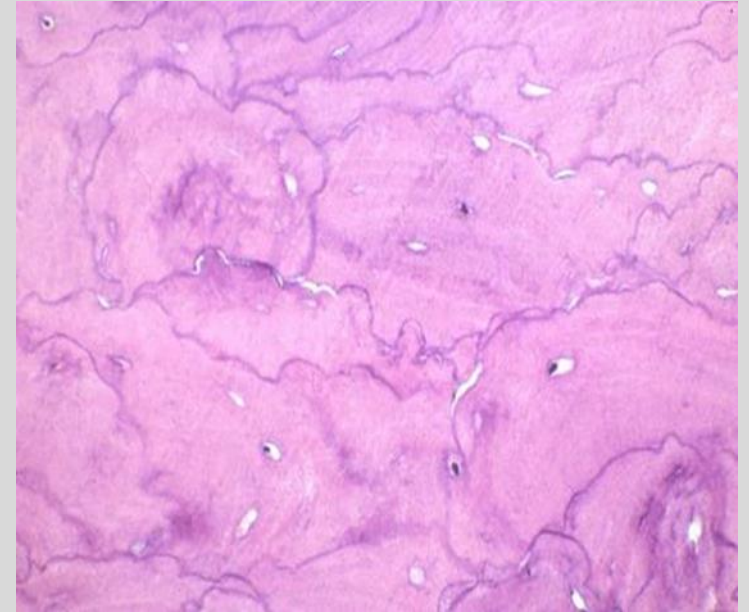


PAGET DISEASE

- **Histology Hallmark** - mosaic pattern of lamellar bone develops in sclerotic phase (jigsaw puzzle–like appearance produced by prominent cement lines, which join haphazardly oriented units of lamellar bone)
- Less specific during other phases of disease
- **Osteoclasts: abnormally large** with increased nuclei; (e.g., 100 nuclei); persist in mixed phase, but many bone surfaces also lined by **plump osteoblasts**
- Marrow adjacent to bone-forming surface replaced by loose connective tissue containing osteoprogenitor cells and blood vessels
- Coarsely thickened trabeculae and soft and porous cortices lack structural stability
- The bone is prone to fractures

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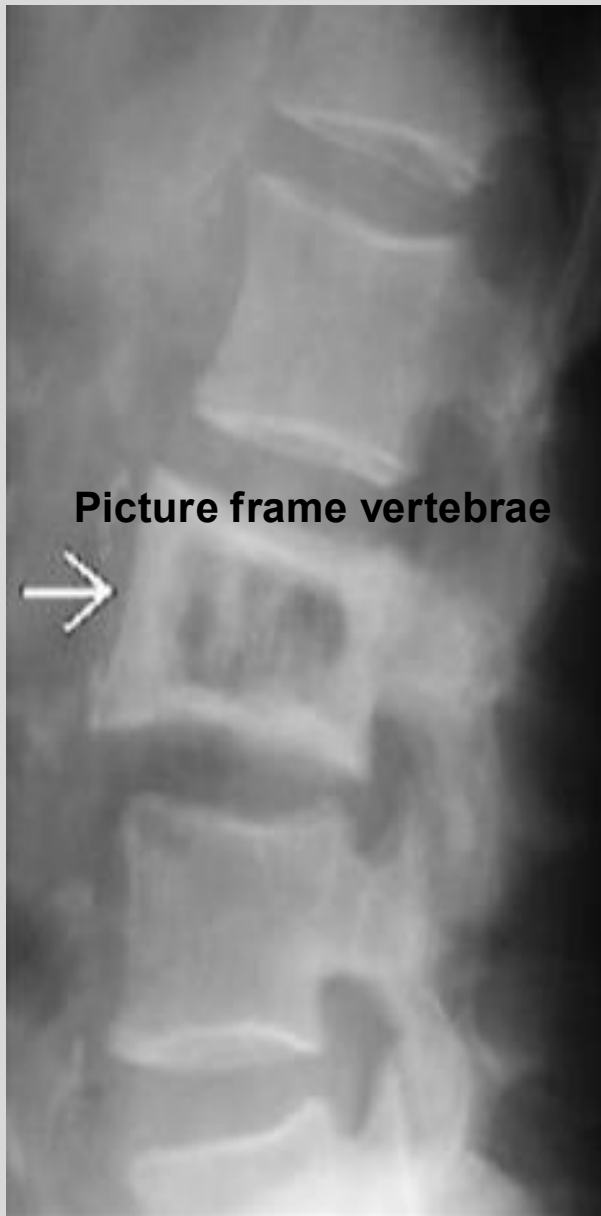
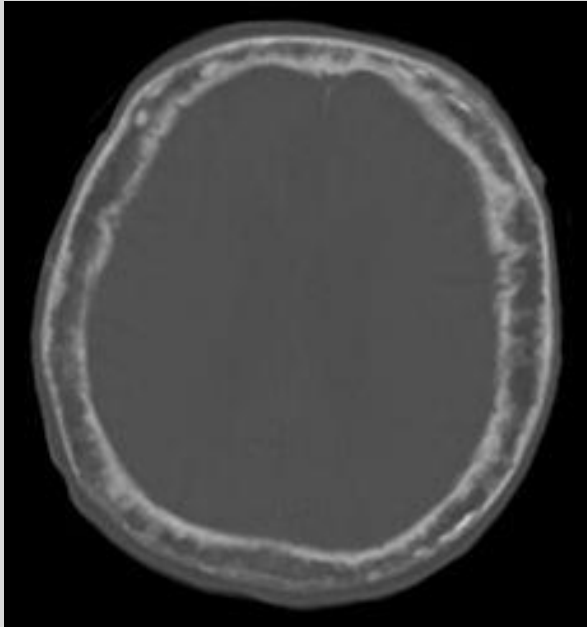
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Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 26

Skull thickening

- May be associated with auditory foramen narrowing which followed by hearing loss



CLINICAL FEATURES

Monostotic: 15% of cases

Polyostotic : 85%;

Overall: axial skeleton or proximal femur involved in up to 80% of cases

- **Most cases asymptomatic, discovered as incidental radiographic finding**

Pain localized to the affected bone may be present due to microfractures or bone overgrowth that compresses spinal and cranial nerve roots

Enlargement of craniofacial skeleton may produce **leontiasis ossea (lion face)** and a heavy cranium: the patient cannot hold head up.

Weight bearing causes **anterior bowing of the femurs and tibiae** and distorts femoral heads; results in severe **secondary osteoarthritis**

- **Chalk stick–type fractures:** common; long bones of lower extremities
- **Vertebral compression fractures:** spinal cord injury and kyphosis
- **High output heart failure: A-V shunts**

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- Rarely, hypervascularity of Pagetic bone warms overlying skin; in severe polyostotic disease increased blood flow can act as an **arteriovenous shunt, leading to high-output heart failure**
- **Sarcoma: less than 1% of all individuals with Paget disease;**
- **Labs: ↑ ALP – Marker for ↑ bone formation; Normal; Calcium, Phosphorus and PTH**
- Usually not serious or life-threatening; mild symptoms
- **Therapy:** calcitonin and bisphosphonates

OSTEONECROSIS-AVASCULAR NECROSIS

- **Most common causes:** fractures and corticosteroid administration

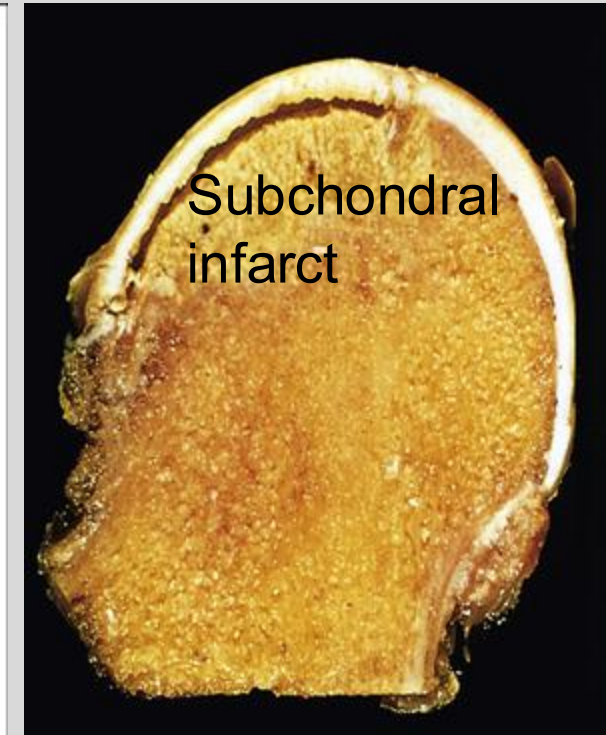
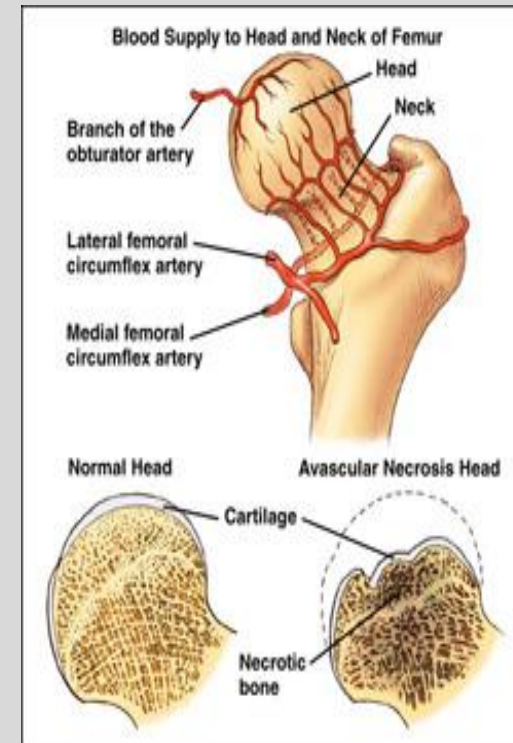
Location: femoral head due to medial circumflex femoral artery; 80% of time bilateral hip involvement

Predisposing conditions: alcohol abuse, bisphosphonate therapy, connective tissue disease, **trauma**, chronic pancreatitis, Gaucher disease, pregnancy, radiation therapy, **sickle cell crisis**, tumors, decompression sickness

- **Medullary infarcts** are geographic in shape; involve both trabecular bone and marrow
- **Subchondral infarcts:** triangular or **wedge-shaped** segment with the subchondral bone plate as base (undergoes necrosis); overlying articular cartilage remains viable due to nutrients within synovial fluid
- **Micro:** empty lacunae surrounded by necrotic adipocytes
- **Symptoms** depend on location and extent of infarct. **Subchondral infarcts** : pain initially associated with activity, decreased ROM; it collapses and results in secondary osteoarthritis

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